

## SEMESTER S7

### REAL TIME SYSTEMS

(Common to CS/CM/CA/AM)

|                                      |                    |                    |                |
|--------------------------------------|--------------------|--------------------|----------------|
| <b>Course Code</b>                   | <b>PECST748</b>    | <b>CIE Marks</b>   | 40             |
| <b>Teaching Hours/Week (L:T:P:R)</b> | 3:0:0:0            | <b>ESE Marks</b>   | 60             |
| <b>Credits</b>                       | 3                  | <b>Exam Hours</b>  | 2 Hrs. 30 Min. |
| <b>Prerequisites (if any)</b>        | PCCST402, PCCST403 | <b>Course Type</b> | Theory         |

#### Course Objectives:

1. To enable the learners to familiarize with the concepts of Real Time systems
2. To teach different task scheduling algorithms in uniprocessor and multiprocessor environments.
3. To learn the features of real-time communications, real-time databases and real time OS.

### SYLLABUS

| <b>Module No.</b> | <b>Syllabus Description</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                       | <b>Contact Hours</b> |
|-------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------|
| <b>1</b>          | Introduction to Real-Time systems: Basic concepts, applications of Real-Time systems, basic model of Real-Time systems, characteristics of Real-Time systems, types of Real-Time systems: hard, firm, soft, timing constraints, modelling timing constraints.                                                                                                                                                                                                                     | <b>6</b>             |
| <b>2</b>          | Real-Time task scheduling: Basic concepts, clock driven scheduling, table driven scheduling, cyclic, schedulers, hybrid schedulers, event driven scheduling, EDF Scheduling, RMA, DMA, resource sharing among RT tasks, Priority inversion, Priority Inheritance Protocol, Highest Locker Protocol, Priority Ceiling Protocol, Scheduling Real-Time tasks in multiprocessor and distributed systems, Fault tolerant scheduling of tasks, clocks in distributed Real-Time systems. | <b>12</b>            |
| <b>3</b>          | Commercial Real-Time Operating Systems: Time services, Features of real-time operating systems, UNIX and Windows as RTOS, POSIX, PSOS, VRTX, QNX, RT Linux, Lynx, other RTOS, benchmarking RT OS, Real-Time OS: OS services, I/O subsystem, Network OS.                                                                                                                                                                                                                           | <b>8</b>             |
| <b>4</b>          | RT communications: QoS framework, models, Real-Time Communication in a LAN, IEEE 802.4, RETHER, Communication over Packet Switched Networks, Routing algorithms, RSVP, rate control; RT databases - Applications, characteristics of temporal data, Concurrency control, Commercial RT databases, Special topics in Real-Time systems.                                                                                                                                            | <b>10</b>            |

**Course Assessment Method**  
(CIE: 40 marks, ESE: 60 marks)

**Continuous Internal Evaluation Marks (CIE):**

| Attendance | Assignment/<br>Microproject | Internal<br>Examination-1<br>(Written) | Internal<br>Examination- 2<br>(Written) | Total |
|------------|-----------------------------|----------------------------------------|-----------------------------------------|-------|
| 5          | 15                          | 10                                     | 10                                      | 40    |

**End Semester Examination Marks (ESE)**

*In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions*

| Part A                                                                                                                                                                                         | Part B                                                                                                                                                                                                                                                                                                      | Total     |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------|
| <ul style="list-style-type: none"> <li>2 Questions from each module.</li> <li>Total of 8 Questions, each carrying 3 marks</li> </ul> <p style="text-align: center;"><b>(8x3 =24 marks)</b></p> | <ul style="list-style-type: none"> <li>Each question carries 9 marks.</li> <li>Two questions will be given from each module, out of which 1 question should be answered.</li> <li>Each question can have a maximum of 3 subdivisions.</li> </ul> <p style="text-align: center;"><b>(4x9 = 36 marks)</b></p> | <b>60</b> |

**Course Outcomes (COs)**

At the end of the course students should be able to:

| Course Outcome |                                                                                                            | Bloom's<br>Knowledge<br>Level (KL) |
|----------------|------------------------------------------------------------------------------------------------------------|------------------------------------|
| CO1            | Explain the various Real Time applications, services, design considerations and architectures              | K2                                 |
| CO2            | Develop efficient algorithms for real-time task scheduling in uniprocessor and multiprocessor environments | K3                                 |
| CO3            | Identify the limitations of a non real-time operating system in running a real-time application            | K2                                 |
| CO4            | Identify and address the important issues in real-time communications                                      | K2                                 |
| CO5            | Understand the concepts of use real-time databases                                                         | K2                                 |

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

**CO-PO Mapping Table (Mapping of Course Outcomes to Program Outcomes)**

|            | PO1 | PO2 | PO3 | PO4 | PO5 | PO6 | PO7 | PO8 | PO9 | PO10 | PO11 | PO12 |
|------------|-----|-----|-----|-----|-----|-----|-----|-----|-----|------|------|------|
| <b>CO1</b> | 3   | 3   |     |     |     |     |     |     |     |      |      | 3    |
| <b>CO2</b> | 3   | 3   | 3   |     |     |     |     |     |     |      |      | 3    |
| <b>CO3</b> | 3   | 3   | 2   |     |     |     |     |     |     |      |      | 3    |
| <b>CO4</b> | 3   | 3   | 2   |     |     |     |     |     |     |      |      | 3    |
| <b>CO5</b> | 3   | 3   | 2   |     |     |     |     |     |     |      |      | 3    |

Note: 1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation

| <b>Text Books</b> |                                        |                             |                              |                         |
|-------------------|----------------------------------------|-----------------------------|------------------------------|-------------------------|
| <b>Sl. No</b>     | <b>Title of the Book</b>               | <b>Name of the Author/s</b> | <b>Name of the Publisher</b> | <b>Edition and Year</b> |
| 1                 | Real-Time Systems: Theory and Practice | Rajib Mall                  | Pearson Education,           | 1/e, 2007               |
| 2                 | Real-Time Systems                      | Jane W. S. Liu              | Pearson Education,           | 3/e, 2009               |

| <b>Reference Books</b> |                                              |                                        |                              |                         |
|------------------------|----------------------------------------------|----------------------------------------|------------------------------|-------------------------|
| <b>Sl. No</b>          | <b>Title of the Book</b>                     | <b>Name of the Author/s</b>            | <b>Name of the Publisher</b> | <b>Edition and Year</b> |
| 1                      | Real-Time Systems Design and Analysis, Wiley | Philip A. Laplante,<br>Seppo J. Ovaska | Wiley                        | 1/e, 2012               |

| <b>Video Links (NPTEL, SWAYAM...)</b> |                                                                                                                           |
|---------------------------------------|---------------------------------------------------------------------------------------------------------------------------|
| <b>Module No.</b>                     | <b>Link ID</b>                                                                                                            |
| <b>1, 2, 3, 4</b>                     | <a href="https://onlinecourses.nptel.ac.in/noc22_cs104/preview">https://onlinecourses.nptel.ac.in/noc22_cs104/preview</a> |